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ROLL NUMBER: 2420030322

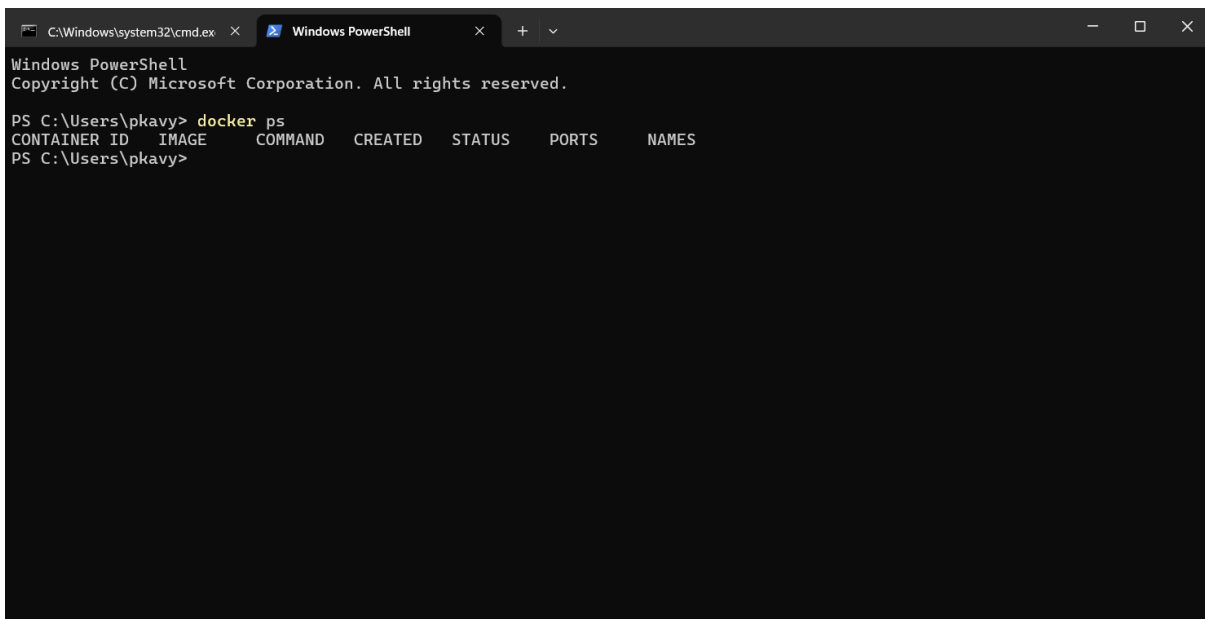
SECTION: 9

ASE TASK-7

KUBERNETES

1) Start docker and minikube

➔ Start the docker desktop



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\pkavy> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\pkavy>
```

➔ Install minikube

- Download minikube from chrome using the link:
https://minikube.sigs.k8s.io/docs/start/?utm_source=chatgpt.com
- Put minikube.exe in C:\minikube
- Open powershell then run

```
$userPath = [Environment]::GetEnvironmentVariable('Path',  
[EnvironmentVariableTarget]::User)  
if ($userPath.Split(';') -notincontains 'C:\minikube') {  
  
    [Environment]::SetEnvironmentVariable('Path',  
"$userPath;C:\minikube", [EnvironmentVariableTarget]::User)  
  
}
```

➔ Check minikube version

minikube version

```
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\pkavy>minikube version
minikube version: v1.39.0
commit: 7a9f6a841470a207de8cf4bafcccee0969d8ba10

C:\Users\pkavy>minikube version
minikube version: v1.39.0
commit: 7a9f6a841470a207de8cf4bafcccee0969d8ba10

C:\Users\pkavy>minikube start --driver=docker
* minikube v1.39.0 on Microsoft Windows 11 Home Single Language 25H2
* Using the docker driver based on user configuration
* Using Docker Desktop driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.51 ...
* Downloading Kubernetes v1.37.0 preload ...
  > gcr.io/k8s-minikube/kicbase: 1.60 MiB / 507.61 MiB [>____] 0.00% ? p/s ?
```

➔ Start Minikube using Docker

minikube start --driver=docker

```
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\pkavy>minikube version
minikube version: v1.39.0
commit: 7a9f6a841470a207de8cf4bafcccee0969d8ba10

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* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.51 ...
* Downloading Kubernetes v1.37.0 preload ...
  > gcr.io/k8s-minikube/kicbase: 1.60 MiB / 507.61 MiB [>____] 0.00% ? p/s ?
```

➔ Check Minikube status

minikube status

```
C:\Windows\system32\cmd.exe: X Windows PowerShell X + v
* minikube v1.39.0 on Microsoft Windows 11 Home Single Language 25H2
* Using the docker driver based on user configuration
* Using Docker Desktop driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.51 ...
* Downloading Kubernetes v1.37.0 preload ...
  > preloaded-images-k8s-v18-v1...: 347.58 MiB / 347.58 MiB 100.00% 1.90 MiB
  > gcr.io/k8s-minikube/kicbase: 507.61 MiB / 507.61 MiB 100.00% 2.53 MiB p
* Creating docker container (CPUs=2, Memory=3500MB) ...
* Preparing Kubernetes v1.37.0 on containerd 2.3.4 ...
* Configuring CNI (Container Networking Interface) ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, default-storageclass

! C:\Program Files\Docker\Docker\resources\bin\kubectl.exe is version 1.34.1, which may have incompatibilities with Kubernetes 1.37.0.
  - Want kubectl v1.37.0? Try 'minikube kubectl -- get pods -A'
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default

C:\Users\pkavy> minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

C:\Users\pkavy>
```

2) Install Kubectl

- ➔ Download from the chrome
- ➔ Verify installation
 - Kubectl version
 - Kubectl get nodes

```
C:\Users\pkavy>kubectl version
Client Version: v1.34.1
Kustomize Version: v5.7.1
Server Version: v1.37.0
Warning: version difference between client (1.34) and server (1.37) exceeds the supported minor version skew of +/-1

C:\Users\pkavy>kubectl get nodes
NAME        STATUS    ROLES    AGE     VERSION
minikube    Ready     control-plane  5m58s   v1.37.0

C:\Users\pkavy>
```

3) Create and deploy nodes

- ➔ Create YAML file
 - Notepad pod.yaml
 - Paste Nginx pod YAML code-> save

```
C:\Windows\system32\cmd.exe x Windows PowerShell x + - □ x
! C:\Program Files\Docker\Docker\resources\bin\kubectl.exe is version 1.34.1
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- Want kubectl v1.37.0? Try 'minikube kubectl -- get pods -A'
* Done! kubectl is now configured to use "minikube" cluster and "default" na
mespace by default

C:\Users\pkavy> minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
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kubeconfig: Configured

C:\Users\pkavy> kubectl version
Client Version: v1.34.1
Kustomize Version: v5.7.1
Server Version: v1.37.0
Warning: version difference between client (1.34) and server (1.37) exceeds
the supported minor version skew of +/-1

C:\Users\pkavy> kubectl get nodes
NAME STATUS ROLES AGE VERSION
minikube Ready control-plane 5m58s v1.37.0

C:\Users\pkavy> cd "C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Lab-6"

C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Lab-6> notepad pod.yaml

C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Lab-6>
```

```
Syl.txt • questi • docx.t • To run • review • sl ► +
File Edit View
apiVersion: v1
kind: Pod
metadata:
  name: nginx
spec:
  containers:
  - name: nginx
    image: nginx
    ports:
    - containerPort: 80

Ln 10, Col 28 146 characters Plain text 100% Windows (CRLF) UTF-8
```

➔ Deploy node

- Kubectl apply -f pod.yaml

➔ Verify pod

- Kubectl get pods
- Kubectl describe pod nginx

```
C:\Windows\system32\cmd.exe x  Windows PowerShell x + v
Customize Version: v5.7.1
Server Version: v1.37.0
Warning: version difference between client (1.34) and server (1.37) exceeds the supported minor version skew of +/-1

C:\Users\pkavy>kubectl get nodes
NAME        STATUS    ROLES    AGE     VERSION
minikube    Ready     control-plane  5m58s   v1.37.0

C:\Users\pkavy>cd "C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Lab-6"

C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Lab-6>notepad pod.yaml

C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Lab-6>dir pod.yaml
Volume in drive C is Windows-SSD
Volume Serial Number is C2F5-29AA

Directory of C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Lab-6

20-09-2026  11:52 AM                155 pod.yaml
             1 File(s)                155 bytes
             0 Dir(s)  256,155,172,864 bytes free

C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Lab-6>kubectl apply -f pod.yaml
pod/nginx created

C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Lab-6>kubectl get pods
NAME        READY   STATUS    RESTARTS   AGE
nginx       0/1     ContainerCreating   0           9s
```

```
nginx       0/1     ContainerCreating   0           9s

C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Lab-6>kubectl describe pod nginx
Name:        nginx
Namespace:   default
Priority:     0
Service Account: default
Node:        minikube/192.168.49.2
Start Time:  Sun, 20 Sep 2026 11:52:24 +0530
Labels:      <none>
Annotations: <none>
Status:      Pending
IP:          <none>
IPs:         <none>
Containers:
  nginx:
    Container ID:
    Image:        nginx
    Image ID:
    Port:         80/TCP
    Host Port:    0/TCP
    States:       Waiting
      Reason: ContainerCreating
    Ready:        False
    Restart Count: 0
    Environment: <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-mjvk8 (ro)
Conditions:
  Type              Status
  PodReadyToStartContainers  True
  Initialized         True
  Ready               False
  ContainersReady     False
  PodScheduled        True
Volumes:
  kube-api-access-mjvk8:
    Type:              Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName:       kube-root-ca.crt
    Optional:            false
```

4)Access pod

➔ Port forward

- Kubectl port -forward nginx 8080:80

```
C:\Windows\system32\cmd.exe x Windows PowerShell x + v
nginx:
  Container ID:
  Image:      nginx
  Image ID:
  Port:      80/TCP
  Host Port: 80/TCP
  State:      Waiting
  Reason:     ContainerCreating
  Ready:      False
  Restart Count: 0
  Environment: <none>
  Mounts:
    /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-mjvk8 (ro)
Conditions:
  Type              Status
  PodReadyToStartContainers True
  Initialized        True
  Ready              False
  ContainersReady    False
  PodScheduled       True
Volumes:
  kube-api-access-mjvk8:
    Type:              Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName:      kube-root-ca.crt
    Optional:           false
    DownwardAPI:        true
QoS Class:           BestEffort
Node-Selectors:      <none>
Tolerations:         node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                     node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type     Reason      Age   From          Message
  ----     -
  Normal   Scheduled   19s   default-scheduler Successfully assigned default/nginx to minikube
  Normal   Pulling     19s   kubelet       Pulling image "nginx"

C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Lab-6>kubectl port-forward nginx 8080:80
Forwarding from 127.0.0.1:8080 -> 80
Forwarding from [::1]:8080 -> 80
```

➔ Open browser

- Visit: <http://localhost:8080>

